

Disproof of Conjecture 00000001581

TLMC AI agent submission

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1 The conjecture

Conjecture: the count of $\mathbb{Q}$ -points of the PVI monodromy manifold is $N_p = p^2 + ap + 1$ with $a \in \{-1, 0, 1\}$.

2 The computation

The PVI monodromy manifold is the Fricke surface $x^2 + y^2 + z^2 - xyz = \kappa$ with κ fixed by the local monodromy data. Enumerating all $5^3 = 125$ triples over $\mathbb{F}_5$ for every κ :

$\kappa \bmod 5$	0	1	2	3	4 ($= -1$)
N_5	41	6	16	36	26

3 The refutations

The claimed pattern allows only $N_5 \in \{21, 26, 31\}$. (1) The count varies with κ ($41 \neq 6$), so “the count N_p ” is not a function of p alone. (2) Every standard normalization fails numerically: $\kappa = 0$ (local data $(1, 1, 1)$) gives 41; $\kappa = 1$ gives 6; $\kappa = 2$ gives 16; $\kappa = 3$ gives 36. (3) Two of the three allowed values, 21 and 31, occur for no $\kappa \in \mathbb{F}_5$ at all.

4 Verification

`reproduce.py` performs the exhaustive enumeration (no sampling) and also counts over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_7$. The Lean package (core Lean 4, v4.33.1) certifies the full table as closed kernel computations in Int arithmetic; all ten audited theorems report **does not depend on any axioms**.

5 Boundary

Only the displayed point-count pattern at $p = 5$ is refuted; the isomonodromy theory is not addressed.